HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed linguistic datasets and audio outputs for Text-to-Speech (TTS) quality, identifying **language-specific challenges** and proposing test scenarios for **Italian** and **Spanish** locales.
- Developed and maintained text normalization test sets and rules, ensuring accuracy against defined test sets and improving **grapheme-to-phoneme (G2P)** evaluation.
- Performed targeted data quality checks and audio evaluations using structured data and text-processing techniques, including **Pandas** and **regular expressions (Regex)**.
- Contributed to improving the TTS pipeline by supporting enhancements in text normalization and pronunciation prediction for **Brazilian Portuguese**.
- Investigated and resolved language-related issues reported through the Linguist Runbook, continuing the improvement of text normalization and linguistic rules.
- Applied computational experience in speech and language technologies to analyse text and linguistic datasets, identifying patterns and opportunities for improvement.
- Managed production environments for enterprise applications at HCL Technologies, focusing on SLA management and P1/P2 incident response using **Splunk** and **Dynatrace**.

SKILLS

Linguistics & TTS – Text Normalization, Pronunciation Prediction, Grapheme-to-Phoneme (G2P) Evaluation, Linguistic Analysis, Audio Quality Evaluation, IPA

Data Analysis & Processing – Pandas, Regular Expressions (Regex), Python, SQL, Data Quality Checks, Dataset Analysis

Programming & Scripting – Python, Java, Spring Boot, REST APIs

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Shell Scripting

Development Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile/Scrum

PROJECTS

TTS Linguistic Quality Assurance Framework

- Designed and implemented a framework for evaluating linguistic and audio quality of Text-to-Speech outputs for **Italian** and **Spanish**, identifying specific phonetic and grammatical challenges.
- Developed Python scripts incorporating **Pandas** and **Regex** to automate the analysis of linguistic datasets, flagging inconsistencies and errors in text normalization rules.
- Created comprehensive test sets for grapheme-to-phoneme (G2P) prediction, ensuring high accuracy and coverage for critical pronunciation scenarios.

Enterprise Application Support & Automation Suite

- Built Python automation scripts for data extraction and reporting from enterprise systems, reducing manual effort by an estimated **15 hours per week** for the operations team.
- Managed production support for Java Spring Boot applications, including incident response and root cause analysis using **Splunk** for log monitoring and troubleshooting.
- Developed and maintained CI/CD pipelines using **Jenkins** for automated builds and deployments, improving deployment consistency across development and production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)